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Inventor(s)

Shepherd; Rebecca et al.

VIAL ADAPTER AND VIAL ADAPTER SYSTEM AND METHOD

Abstract

A vial adapter may include a body extending from a proximal end to a distal end. A syringe connector configured to engage a syringe may be proximate the proximal end of the body. A cannula configured to pierce a septum of a vial may be disposed proximate the distal end of the body. A vial attachment portion may be disposed at the distal end of the body. A fluid conduit may extend from an outlet aperture at the syringe connector to the cannula. A compressible member may be disposed within the syringe connector. The syringe connector may axially compress the compressible member from an expanded configuration, in which the compressible member blocks the outlet aperture and prevents fluid flow from the fluid conduit to the syringe connector, to a compressed configuration, in which the outlet aperture is exposed.

Inventors:	Shepherd; Rebecca (Runcorn Cheshire, GB), Jurilj; Ivan (Zagreb, HR), Byrne; Oliver (Runcorn Cheshire, GB), Bingula; Silvestar (Zagreb, HR)
Applicant:	NORTON HEALTHCARE LIMITED (Castleford, West Yorkshire, GB)
Family ID:	86185060
Assignee:	NORTON HEALTHCARE LIMITED (Castleford, West Yorkshire, GB)
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. provisional patent application No. 63/333,623, filed Apr. 22, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] In a medical context, connection between a syringe containing a medical substance or to be filled with a medical substance and a vial containing another medical substance is often required, for example for safety reasons, to ensure a controlled transfer of fluids, and/or mixing/reconstitution of materials just prior to administration. Furthermore, certain vial adapters allow for plural syringe attachments (e.g., whether the same or different syringe) for multiple dose applications.

[0003] Existing solutions for connecting the syringe to the vial include different vial-syringe connection devices such as syringe adaptors and vial adaptors. One such solution is described, for example, in commonly owned PCT Publication WO 2022/049110A1, published Mar. 10, 2022, and entitled Vial Adapter with Valve, the entire contents of which are incorporated herein by reference.

[0004] Within this context, there is a need for an improved solution for connecting a vial and a syringe, including limiting access, where desired, to medicament.

SUMMARY

[0005] The application is generally related to devices and methods used for handling and manipulating fluids in a medical context, and more particularly, to vial adapters and vial adapter systems and related methods.

[0006] According to an example embodiment of the present invention, vial adapters, systems and methods may include a vial adapter configured to limit the number of times a syringe can be used with an adapter. In some examples, the vial adapters, systems and methods may additionally, or alternatively, limit the medicament dose accessible during any given syringe use.

[0007] In one example, a vial adapter may include one or more of a body extending from a proximal end to a distal end, a syringe connection at the proximal end of the body, the syringe connection configured to engage a syringe, and a cannula disposed at the distal end of the body. The cannula may have a pointed tip, for example so as to pierce a septum of a vial. The syringe connection may be a syringe connector. The vial adapter may include a vial attachment portion at the distal end of the body. The vial attachment portion may be configured to attach to the vial. The vial adapter may include a fluid conduit. The fluid conduit may extend from an outlet aperture at the syringe connector to the cannula. In some examples, the vial attachment may include a compressible member. The compressible member may be disposed within the syringe connector. Attachment of a syringe to the syringe connector may (e.g., axially) compress the compressible member. For example, the compressible member may be compressed from an expanded configuration to a compressed configuration. In the expanded configuration, the compressible member may block the outlet aperture and/or prevent fluid flow from the fluid conduit to the syringe connector. In the compressed configuration the outlet aperture may be exposed and/or the

fluid conduit may be in fluid communication with the syringe connector.

[0008] In some examples, a vial adapter may include a body. The body may include (e.g., define) a proximal end and a distal end. A fluid conduit may extend through the body, for example from the proximal end to the distal end. The body may include (e.g., define) a circumferential wall at a distal end of the conduit. A cannula may be disposed proximal (e.g., at) the distal end of the body. The cannula may include a pointed tip, for example to puncture a septum of a vial. The vial adapter may additionally, or alternatively, include a vial attachment portion. The vial attachment portion may be disposed about the fluid conduit and/or configured to engage the vial. The vial adapter may include a gasket. The gasket may be disposed within and/or configured to move axially through the fluid conduit. The gasket may include a syringe connector. The syringe connector may be configured to engage a syringe and/or at least one aperture opposite from and in fluid communication with the syringe connector. The aperture may be radially offset from a center of the gasket. The gasket may be movable through the fluid conduit, for example to a distalmost position. In the distalmost position, the gasket may engage the circumferential wall, for example to block the aperture and/or prevent fluid flow from the cannula to the syringe connector portion.

[0009] In some examples, a vial adapter may include a main body. The main body may include a vial attachment portion, for example configured to engage a vial. Additionally, or alternatively, the main body may include a port, for example defining an internal thread. The vial adapter may include an attachment portion. The attachment portion may include a cannula, for example configured to extend through the main body and at least partially into the vial. Additionally, or alternatively, the attachment portion may include an external thread. The external thread may be complementary to (e.g., engage) the internal thread of the port of the main body, such that engagement of the external thread and the internal thread secures the attachment portion to the main body. Incremental advancement of the external thread along the internal thread via rotation of the attachment portion may adjust a distance the cannula extends into the vial.

[0010] Additional features and advantages are realized through the system of the present invention. Other embodiments and aspects of the invention are described in detail herein and are considered a part of the claimed invention. For a better understanding of the invention with advantages and features, refer to the description and to the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Many aspects of the invention can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale, with emphasis being placed upon clearly illustrating the principles of the present invention. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views. Furthermore, each drawing contained in this provisional application includes at least a brief description thereon and associated text labels further describing associated details.

[0012] FIG. 1 is a schematic view of an example of a syringe and vial adapter;

[0013] FIG. 2A is a schematic view of a cam track for a vial adapter in accordance with the vial adapter of FIG. 1;

[0014] FIG. 2B is a close-up of a portion of the vial adapter cam track of FIG. 2A;

[0015] FIG. 3A is a first schematic image of an example ratchet vial adapter;

[0016] FIG. 3B is another schematic image of the example ratchet vial adapter of FIG. 3A;

[0017] FIG. 3C is another schematic image of the example ratchet vial adapter of FIG. 3A;

[0018] FIG. 3D is another schematic image of the example ratchet vial adapter of FIG. 3A;

[0019] FIG. 3E is another schematic image of the example ratchet vial adapter of FIG. 3A;

[0020] FIG. 4 is a diagram view of an example vial and vial adapter;

[0021] FIG. 5A is a cross sectional view an example vial and vial adapter;
[0022] FIG. 5B is a cross sectional view an example vial and vial adapter, in accordance with FIG. 5A;
[0023] FIG. 5C is a cross sectional view an example vial and vial adapter, in accordance with FIG. 5A;
[0024] FIG. 5D is a cross sectional view an example vial and vial adapter, in accordance with FIG. 5A;
[0025] FIG. 6A is a perspective view an example vial adapter including a dose/cannula setting;
[0026] FIG. 6B is a diagram view the example vial adapter of FIG. 6A;
[0027] FIG. 6C are cross sectional views of the example dose setting aspects of FIGS. 6A and 6B;
[0028] FIG. 7A is a first schematic image of an example ratchet vial adapter;
[0029] FIG. 7B is another schematic image of the example ratchet vial adapter of FIG. 7A;
[0030] FIG. 7C is another schematic image of the example ratchet vial adapter of FIG. 7A;
[0031] FIG. 7D is another schematic image of the example ratchet vial adapter of FIG. 7A;
[0032] FIG. 7E is another schematic image of the example ratchet vial adapter of FIG. 7A;
[0033] FIG. 7F is another schematic image of the example ratchet vial adapter of FIG. 7A; and
[0034] FIG. 7G is another schematic image of the example ratchet vial adapter of FIG. 7A.

DETAILED DESCRIPTION OF THE INVENTION

[0035] In furtherance of the brief description provided above and associated textual detail of each of the figures, the following description provides additional details of example embodiments of the present invention.

[0036] Detailed illustrative examples are disclosed herein. However, specific functional details disclosed herein are merely representative for purposes of describing example embodiments. Examples may, however, be embodied in many alternate forms and should not be construed as limited to only the embodiments set forth herein.

[0037] Accordingly, while example embodiments are capable of various modifications and alternative forms, embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms disclosed. To the contrary, example embodiments are to cover all modifications, equivalents, and alternatives falling within the scope of example embodiments.

[0038] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, steps, and/or calculations, these elements, steps, and/or calculations should not be limited by these terms. These terms are only used to distinguish one element, step, and/or calculation from another. For example, a first calculation could be termed a second calculation, and, similarly, a second step could be termed a first step, without departing from the scope of this disclosure. As used herein, the term “and/or” and the “/” symbol includes any and all combinations of one or more of the associated listed items.

[0039] As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises”, “comprising”, “includes” and/or “including”, when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Therefore, the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of example embodiments.

[0040] It should also be noted that in some alternative implementations, the functions/acts noted may occur out of the order noted in the figures. For example, two figures shown in succession may in fact be executed substantially concurrently or may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

[0041] Herein, example embodiments of the present invention will be described in detail. Example embodiments of the present invention provide a vial adapter, system, and method as described below.

[0042] FIG. 1 illustrates an example vial adapter, shown generally at **100**, along with a syringe **110** configured to connect to the vial adapter **100**. The vial adapter **100** has a body, or casing, **112**, that extends from a proximal end **112a** to a distal end **112b**. The vial adapter **100** may include a conduit **126**. The conduit may extend through the body **112** from the proximal end **112a** to the distal end **112b**. Additionally, or alternatively, the vial adapter may include a cannula **114** and vial attachment portion **120** at the distal end **112b**. In one example, the vial attachment portion **120** can include flexible arms **121**, **122** laterally disposed on opposite sides of the cannula **114**. Additionally, or alternatively, the vial attachment portion **120** may comprise a flexible skirt extending circumferentially about the cannula **114**. The cannula **114** may be disposed at the distal end of the body **112b** and/or configured with a pointed tip so as to puncture the septum of a vial (not shown in FIG. 1 but illustrated in subsequent drawings). Though depicted as a plastic spike, in some examples the cannula **114** may comprise a metal needle or other fluid conduit as desired. The syringe **110** may include connector **116**, for example complementary to proximal syringe connector **118** on the gasket **136**, which is described herein. The connector **116** may be a twist on connector. The vial attachment portion **120** may be disposed on the vial adapter **100**, for example opposite the syringe connector **118**. We note that the illustrated twist on connector is merely an example, and may include one or more of a snap on connection, a luer connection, an integral connection, etc., without limitation.

[0043] Referring still to FIG. 1, the illustrated vial adapter **100** includes a vial attachment portion **120** at the distal end **112b** of the body **112**. The vial attachment portion may be configured to hold a vial (e.g., vial **550** shown in FIG. 5A, for example). The vial may contain medicament. The vial may be in engagement with the vial adapter **100**. In some examples, the arms **121**, **122** are resilient and/or snap onto the vial when the vial is pressed against the vial adapter **100**. Additionally, or alternatively, the arms and/or vial may incorporate locking features, for example preventing flexing of the arms or otherwise preventing removal of the vial after attachment. Further, in some examples, the arms may have other shapes than illustrated or attach in other ways, such as being partially threaded for easy connection, having a snap or luer connection, etc.

[0044] Referring again to the cannula **114**, arrow **124** shows a direction of fluid withdrawal, for example subsequent to connection of a syringe to the vial adapter **100** and/or application of vacuum from the syringe. Alternatively, or additionally, medicament may be first provided to the vial via a first syringe attached to the vial adapter **100**. As such, arrow **124** may show a direction of fluid withdrawal via a second syringe attached to the vial adapter **100**, subsequent to inversion of the second syringe, vial, and vial adapter assembly by a user. Fluid flow may continue from the vial into the conduit **126** of the vial adapter **100** and/or further through apertures **128**, **130** in gasket **136**, for example according to fluid flow directional arrows **132**, **134**. The gasket may include silicone. The body **112** may include a first material and/or the gasket **136** may include at least one second material. The second material may be different than the first material. In some examples, the first material may be a plastic and/or the second material may be a silicone, but other examples are contemplated. The gasket **136** may be positioned within the conduit **126** and/or be configured to move axially within the conduit **126**, for example while maintaining a fluid seal against the walls of the conduit **126**. The gasket **136** may be biased away from the cannula **114** (e.g., towards the proximal end **112a** of the body **112**), for example by a spring **144**. The spring **144** may extend through the conduit **126** and/or be anchored at the distal end **112b** of the body **112**. The spring **144** will be described further below. The conduit **126** may extend through the body **112** of the vial adapter **100** and/or fluidly connect the syringe connector **118** and the cannula **114**. Apertures **128**, **130** positioned at the distal end of the gasket **136** allow passage of fluid from the conduit **126** to a proximal fluid conduit space **138**, for example within gasket **136** that may feed the fluid through

proximal syringe connector **118** and/or ultimately into the syringe **110**. Though gasket **136** is shown with two apertures **128**, **130**, the gasket **136** may include a single aperture, for example extending circumferentially about the distal end of the gasket **136** to provide access to conduit space **138**. Alternatively, or additionally, the gasket **136** may include more than two apertures (e.g., three, four, five, or more apertures). As shown, the apertures **128**, **130** may be radially offset from the center of the gasket **136** (e.g., and likewise radially offset from the cannula **114**). The apertures **128**, **130** may be axially aligned with a circumferential wall **129** at the distal end of the conduit **126** proximal to the cannula **114**.

[0045] In some examples, the gasket **136** may have a monolithic body defined by one material, for example silicone. The gasket **136** may have a two-part construction (e.g., as illustrated), for example with a proximal part **136a** of the gasket **136** being defined by a first material, such as plastic or silicone, and/or a distal part **136b** of the gasket **136** being defined by a second material, such as a silicone or rubber. In some examples, the second material may be specifically configured (e.g., selected) for providing a fluid seal against the circumferential wall **129** (e.g., as will be described further herein).

[0046] In some examples, the vial adapter **100** may include a ratchet style cam track **140** disposed within the conduit **126**. The gasket **136** may include one or more cam members **142** that can move within the cam track **140**. The vial adapter **100** may include a spring **144** that may bias the gasket **136** away from the cannula **114**. In some examples, the spring **144** is a coil spring, though other types of springs are contemplated. Alternatively, or additionally, the spring **144** may be positioned coaxially with the cannula **114** (e.g., radially spaced from the apertures **128**, **130**). As will be described in more detail below (e.g., and as shown by subsequent cam member positions in FIG. **1**), in some examples, adding successive syringes may advance the cam members **142** of the gasket **136** along the cam track **140**. In some examples, the gasket **136** and cam members **142** may not (e.g., cannot) be advanced to the next position until a syringe **110** is disconnected from the gasket **136** (e.g., through withdrawal of a dose of medicament and removal of the syringe). With the last syringe permitted, the cam members **142** may bottom out within the conduit **126**. In this distalmost position of the gasket **136**, the gasket **136** may engage the circumferential wall **129**, for example to block the apertures **128**, **130** and/or prevent fluid flow from the cannula **114** to the syringe connector **118**. The gasket **136** may comprise more than one material. For example, in the distalmost position the distal portion **136b** may engage the circumferential wall **129** (e.g., when the gasket comprises more than one material). The illustrated example allows fluid transfer between the syringe and the vial via two successive syringes and/or blocks an attempt to perform a fluid transfer via a third syringe. However, it is also contemplated that the vial adapter **100** can be configured to only allow fluid transfer via a single syringe. Also, it is contemplated that the vial adapter **100** can be configured to allow withdrawal of three or more (e.g., any number of) doses.

[0047] FIGS. **2A** and **2B** show an example cam track **140** in more detail (e.g., noting that a single cam track or plural cam tracks may be utilized). With regard to FIG. **2A**, cam track **140** is shown with cam member **142** in successive positions **142A-F** (e.g., which reflect the corresponding positions of the gasket **136** as it moves through the conduit **126**). At positions **142B**, **E** and **F**, a lodging member **210** disposed within the cam track **140** may provide a temporary stop (e.g., during dose transfer and until removal of the syringe after transfer of a dose/removal of the syringe). Once the syringe **110** is disconnected from the gasket **136**, the cam member(s) may move to positions **142 C**, **E**, for example to await attachment of the next syringe. FIG. **2B** provides a close-up of the position **142D**, showing the cam track **140**, the movement of the cam member into position **142D** at arrow **214**, and subsequent movement of the cam member **142** over the lodging member **210** after removal of the syringe **110** at arrow **216** as the cam member **142** moves to position **142E** (e.g., as shown in FIG. **2A**). In some examples, successive movements for one or more advancements results in audible click(s) and/or tactile feedback.

[0048] The cam track **140** may include a one-way stop **205** immediately preceding position **142D**.

In some examples, the one-way stop **205** comprises a ledge, though other one-way stops are contemplated, such as deflectable latches, etc. The one-way stop **205** may allow the cam member **142** to easily transition from position **142C** to **142D** and/or (e.g., but) prevent reverse movement back to position **142C** from position **142D**. The one-way stop **205** may also provide a tactile and/or audial feedback to the user via interaction with the cam member **142**. Though depicted immediately preceding position **142D**, a similar one-way stop **205** may also be disposed immediately preceding position **142B** and/or position **142F**. The cam track **140** may additionally, or alternatively include a (e.g., further) one-way stop **215** (e.g., immediately) following position **142D**, for example adjacent to lodging member **210**. Like one-way stop **205**, one-way stop **215** may comprise a ledge (e.g., though in other embodiments could be differently configured, for example as a one-way latch). The one-way stop **215** may allow the cam member **142** to easily transition out of position **142D** and/or (e.g., but) prevent reverse movement back to position **142D**. The one-way stop **215** may additionally, or alternatively provide a tactile and/or audial feedback to the user via interaction with the cam member **142**. Though depicted immediately following position **142D**, a similar one-way stop **214** may also be disposed (e.g., immediately) following position **142B**.

[0049] With further reference to FIG. 2A, a method of utilizing the vial adapter with a cam track **140** is described. At position **142A**, a first syringe is attached to the adapter. As the syringe is pushed or screwed onto the adapter and/or pressure is applied, the cam member **142** (e.g., which may be a sphere or other shape in the cam track) may move a (e.g., limited and/or predetermined) distance to position **142B**, at which point it may be stopped by lodging member **210** for example. As the first syringe is disengaged from the gasket **136**, the cam member **142** may move over lodging member **210** to position **142C** at least partially under the force of spring **144**. When a second syringe is attached and/or pressure is applied, the cam member **142** may advance through the cam track **140** and/or lodge at position **142D** (also shown in FIG. 2B), for example against a subsequent lodging member **210**. As the second syringe is disengaged, the cam member **142** may move over the lodging member **210** at least partially under the force of spring **144** to position **142E**. If a third syringe is attached and pressure is applied, the cam member may move to position **142F**. However, the third syringe may bottom out the gasket **136** in the cam track **140** at that point, for example via engagement between the gasket **136** and the wall **129**. Alternatively, or additionally, the flow channel may close, for example as the apertures **128**, **130** align with and/or are blocked by the wall **129**, thus preventing flow of medicament from the cannula **114** to the syringe.

[0050] FIGS. 3A-E illustrate a vial adapter **300**. Each (e.g., different) figure illustrates a different state of operation of the vial adapter **300**. FIG. 3A illustrates the vial adapter state prior to attaching a syringe (e.g., syringe **110**). The vial adapter **300** may include a body **312** that extends from a proximal end **312a** to a distal end **312b**. The vial adapter **300** may include one or more of a syringe connector **310** at the proximal end **312a** of the body, a fluid conduit **314** that extends from the proximal end **312a** to the distal end **312b** of the body, a cannula **316** at the distal end **312b** of the body **312**, and a vial attachment portion **318** at the distal end **312b** of the body **312** for attaching the vial adapter **300** to a vial (e.g., such as vial **550** shown in FIG. 5A, though not shown in FIGS. 3A-3E). The cannula **316** may be configured with a pointed tip so as to pierce a portion (e.g., a septum) of a vial (e.g., as shown in subsequent figures). Though depicted as a plastic spike, in other embodiments the cannula **316** may comprise a metal needle or other fluid conduit as desired. Unlike the arms illustrated as the vial attachment portions in FIG. 1, this vial attachment is illustrated as a flexible skirt that extends circumferentially about the cannula **316** and receives the vial (e.g., as a snap fit). However, it is contemplated that in other examples the vial attachment portion **318** may be configured as a screw or interference fit. As with the examples of FIG. 1, additional mechanisms for preventing removal of the vial may be included (e.g., as locking tabs, or by virtue of an interference fit itself, allowing one way attachment, but preventing removal).

[0051] Referring still to FIG. 3A, a first compressible member **320** may be attached to a movable

spring **324**, for example extending distally into an open cam space **326** that contains a ratcheted slider **328**. The ratcheted slider **328** may be disposed within the cam space **326** and/or be axially movable through the cam space **326**. The cam space **326** may be disposed axially between the syringe connector **310** and the cannula **316**. As depicted, the compressible member **320** may be configured as a compressible bellows configured to form a fluid seal with the walls of the syringe connector **310** throughout compressive movement. The compressible member **320** may be positioned within the syringe connector **310**. The proximal end of the movable spring **324** may be attached to the compressible member **320** and/or extend distally into the cam space **326**. The movable spring **324** may be attached to the compressible member **320**, such that distal compression of the compressible member **320** causes the movable spring **324** to translate distally through the cam space **326** for example. Likewise, proximal expansion of the compressible member **320** may cause the movable spring **324** to translate proximally through the cam space **326**. The distal end of the movable spring **324** may be configured to engage the ratcheted slider **328**. For example, the distal end of the movable spring **324** may be configured to engage teeth **328a** of the ratcheted slider **328**. Downward movement of the movable spring **324**, for example via engagement between the movable spring **324** and a lateral surface of a tooth **328a** of the ratcheted slider **328**, (e.g., likewise) may force the ratcheted slider **328** to translate downwards within the cam space **326**. However, (e.g., subsequent) upward movement of the movable spring **324**, for example via upward expansion of the compressible member **320**, may cause the distal end of the movable spring **324** to deflect laterally and/or slide over the angled surface of a tooth **328a** of the ratcheted slider **328**, which may result in the ratcheted slider **328** remaining in its axial position. The distal end of the movable spring **324** may deflect towards the ratcheted slider **328** and/or engage the lateral surface of a tooth **328a** proximal to the tooth **328a** with which it was previously engaged, for example upon full expansion of the compressible member **320**.

[0052] The vial adapter **300** may additionally, or alternatively include a fixed spring **330**, for example to prevent reverse movement of the ratcheted slider **328** upon expansion of the compressible member **320** following ratcheted slider **328** downward movement. The fixed spring **330** may be disposed within the cam space **326** and/or attached to the body **312**. For example, a proximal end of the fixed spring **330** may be fixedly secured to a wall of the cam space **326**. When the movable spring **324** moves distally to translate the ratcheted slider **328** distally, the distal end of the fixed spring **330** may deflect laterally, for example to slide over an angled surface of a tooth **328a** of the ratcheted slider **328** and/or (e.g., subsequently) deflect towards the ratcheted slider to engage a lateral surface of a subsequent tooth **328a**. This engagement between the fixed spring **330** and the ratcheted slider **328** may hold the ratcheted slider **328** axially in place during upward movement of the movable spring **324**, for example preventing proximal movement of the ratcheted slider **328**. That the movable spring **324** and/or fixed spring **330** may be elongate pieces metal or another material that may provide sufficient push and hold for the illustrated ratcheted slider **328**, for example while still allowing for lateral deflection via deformation.

[0053] As depicted, the ratcheted slider **328** may define five ratchet positions for the movable spring **324** and/or the fixed spring **330** to engage the ratcheted slider **328**. Each discrete ratchet position may include (e.g., be defined by) a distinct tooth **328a** of the ratcheted slider **328**. For example, a tooth **328a** may include a lateral surface **328b** of a tooth **328a** of the ratcheted slider **328**. The teeth **328a** of the ratcheted slider may be positioned axially along the ratcheted slider **328**. In one example, the ratcheted slider may include (e.g., may define) one or more of, from distal-most position to proximal-most position, a lowest ratchet position **332**, a second lowest ratchet position **334**, a third ratchet position **340**, a second ratchet position **338**, and a first ratchet position **342**. However, it is contemplated that in other embodiments the ratcheted slider **328** can have more or less ratchet positions than shown presently, for example as dictated by the intended number of syringe connections for a particular implementation.

[0054] FIG. 3B illustrates the adapter **300** in a state with a syringe (e.g., syringe **110** in FIG. 1)

attached (e.g., though for convenience, the syringe is not illustrated, though the compressible member **320** is shown compressed as if the syringe connector were compressing it). The attachment may be by one or more of interference fit, snap on, screw on, etc., for example between the syringe and the syringe connector **310** at this proximal portion. The vial adapter **300** may include a fluid conduit **314**, for example that fluidly connects the syringe connector **310** to the cannula **316**. The fluid conduit **314** may include one or more of an output aperture **314a** open to the syringe connector **310**, a lateral extension **314b** extending from the output aperture **314a** to the axial extension **314c**, an axial extension **314c**, and an input extension **314d** that connects the cannula **316** to the axial extension. The axial extension **314c** of the fluid conduit **314** may be laterally offset from the cam space **326** and/or the cannula **316**, for example where the axial extension **314c** comprises a majority of the length of the fluid conduit **314**. However, it is contemplated that in other embodiments, the cannula **316** (e.g., along with vial attachment portion **318**) may be repositioned, for example so as to be coaxial with the cam space **326** and/or the axial extension **314c** of the fluid conduit **314**. In a first (e.g., initial) position, the compressible member **320** may block the output aperture **314a**, thus for example preventing fluid flow from the vial and cannula **316** and out of the syringe connector **310**. However, connection of a syringe to the syringe connector **310** may compress the compressible member **320** distally such that, for example the output aperture **314a** is exposed and in fluid communication with the syringe. Fluid may be (e.g., therefore) permitted to flow from the vial, through the cannula **316** and fluid conduit **314**, and/or through syringe connector **310** to the syringe. The compressible member **320** may expand to its first (e.g., initial) position, thus again blocking the output aperture **314a**, for example upon syringe removal from the vial adapter **300**.

[0055] As shown in FIG. 3A, in a first (e.g., initial) state with the compressible member **320** in an expanded state, the movable spring **324** may engage the ratcheted slider **328** in the third ratchet position **340**, for example while the fixed spring **330** engages the ratcheted slider **328** in the lowest ratchet position **332**. As shown in FIG. 3B, attachment of a first syringe to the vial adapter **300** may cause the compressible member **320** to compress, for example moving the ratcheted slider **328** distally through displacement of the movable spring **324**. This may cause the fixed spring **330** to deflect laterally, for example over the bottommost tooth **328a** of the ratcheted slider **328** and/or move from the lowest ratchet position **332** to the second lowest ratchet position **334**. As stated previously, in this position the compressible member **320** may be compressed such that the output aperture **314a** is exposed to the syringe connector **310** and/or the first syringe, for example allowing for fluid flow through the fluid conduit **314** between the vial and the first syringe. FIG. 3C shows a third configuration of the vial adapter, with the first syringe removed. As noted, compressible member **320** may be in an expanded state (e.g., no syringe attached, and in some examples blocking fluid flow through fluid conduit **314** via output aperture **314a**). In this state, the fixed spring **330** may be (e.g., still) in the second lowest ratchet position **334**, but, for example the movable spring **324** may be retracted to the second ratchet position **338** from the third ratchet position **340**. The movable spring **324** may be retracted to the second ratchet position **338** from the third ratchet position **340** by virtue of the holding power of fixed spring **330** on the ratcheted slider **328**.

[0056] FIG. 3D shows a fourth position of the vial adapter **300**, with a second syringe attached (as with FIG. 3B, above) to the syringe connector **310**. Attachment of the second syringe may (e.g., further) advance the ratcheted slider **328** via downward movement of the movable spring **324**, for example such that the fixed spring **330** moves into the third ratchet position **340** of the ratcheted slider **334**. Additionally, or alternatively, (e.g., as herein), in some examples, compression of the compressible member **320** may unblock the output aperture **314a** and/or allow fluid flow through the fluid conduit **314** between the second syringe and the vial.

[0057] FIG. 3E illustrates an additional state after removal of the second syringe from the vial adapter **300**. Fixed spring **330** may remain at the third ratchet position **340**, and/or the movable

spring may (e.g., now) be at the first ratchet position **342**. Additionally, or alternatively, as with FIG. **3C**, compressible member (e.g., and/or safety valve) **320** may be in an expanded state (e.g., no syringe attached, and in some examples blocking fluid flow via output aperture **314a**). At this point, the vial adapter **300** may be prevented from use with a third syringe, for example since in this position the distal end of the ratcheted slider **328** engages a distal end of the cam space **326** which may prevent further downward movement of the ratcheted slider **328**. As a result, the movable spring **324** may be prevented from moving distally via engagement with the ratcheted slider **328**. The compressible member **320** (e.g., likewise) may be prevented from compressing, and/or may (e.g., permanently) block the outlet aperture **314a** and fluid flow to the syringe connector **310** from the fluid conduit **314**.

[0058] With further reference to FIGS. **3A-E**, an example method of using the vial adapter **300** may include attaching a first syringe, for example resulting in moving the compressible member **320** down. Additionally, or alternatively, attaching a first syringe may result in the ratcheted slider **328** pushing the fixed spring **330** into a locked position during use (e.g., as at FIGS. **3A-3B**, into locked position at **334**). This may (e.g., then) open up the fluid conduit, for example for transfer of fluid from the prefilled syringe (PFS) and to the vial through the fluid conduit **314**. At FIG. **3C**, the first syringe may be removed, and the movable spring **324** may slide up to the second ratchet position **338** (e.g., due to the lack of presence of a syringe). At FIG. **3D**, a second syringe may be inserted, for example causing sliding of the ratchet **328** downwards and/or (e.g., also) moving the fixed spring **330** to the third ratchet position **340**. Alternatively, or additionally, insertion of a second syringe may also result in opening the fluid flow through conduit **314**. At this point, medicament (e.g., a reconstituted drug comprising the mixed components of the PFS and the vial) may be withdrawn from the vial into the second syringe, and/or removed from the vial adapter **300** (e.g., resulting in the configuration shown in FIG. **3E**). At FIG. **3E**, the vial adapter **300** may be locked out by virtue of the position of the ratcheted slider **328** within the cam space **326** and/or may not accept a third syringe in some examples.

[0059] FIG. **4** illustrates another example of a vial adapter **400**, for example with a predetermined or settable cannula length to allow for accurate dose amounts. The vial adapter **400** may include a body **402** that, for example may include a screw type syringe attachment **410** for receiving a syringe, though as with the above, any manner of attachment is contemplated, (e.g., luer, snap on, etc., including additional intermediate attachments, as desired). The vial adapter **400** may (e.g., also) include a cannula **412** extending from the body **402**. The cannula **412** may include a pointed tip, for example to pierce the septum of and at least partially extend into a vial, as shown in FIG. **4**. In this illustrated embodiment, various dose levels and/or cannula **412** lengths are contemplated, for example accounting for overfill **414** around the cannula **412** intrusion into the vial **416**. Example doses are marked off in FIG. **4**, accounting for anticipated air/gas **418**, and/or a range of doses from 0.0 milliliters (ml) (**420**), to 0.9 ml (**422**), to 1.2 ml (**424**) to 1.7 ml (**426**). In some examples, the cannula length may be set to a particular dose withdrawal, and/or may be set at a minimum set distance or within a range with a minimum distance, for example that will ensure that overfill **414** cannot be withdrawn by a user. Though depicted as a plastic spike, in other embodiments the cannula **412** may comprise a metal needle and/or other fluid conduit as desired. The vial adapter **400** may (e.g., also) include a vial attachment portion **404**, for example for attaching the vial adapter **400** to a vial. The vial attachment portion **404** may include one or more (e.g., a plurality of) snap-fit features for engaging the vial, though other examples are contemplated.

[0060] With further reference to FIG. **4**, in some examples allowing for setting of a dose, (e.g., as with an insulin or other injection pen when setting a dose), a rotation and/or other mechanism may be used to adjust the dose, for example with a first dose being 1.7 ml (**426** in FIG. **4**, being the lowest length of the cannula **412**). Other dose options may include 1.2 ml (**424**) or 0.9 ml (**422**), settable (e.g., via rotation of a vial adapter screw or other mechanism), for example with the

cannula being higher for those doses. In some examples, adjustment (e.g., rotation) may only be permitted in a single direction to prevent reverse changes to the length of the cannula for dosing. In additional examples, the adjustment mechanism and/or adapter may have markings indicating dose levels.

[0061] FIGS. 5A-D show vial adapter **500** and vial **550**, (e.g., similar to the embodiment described with regard to FIG. 4) with further examples of settable cannulas. FIG. 5A shows a vial **500** having a two part construction including a main body **502** and attachment **524**. The main body **502** may include a vial attachment portion **514**, for example for connecting the vial adapter **500** to the vial **550**. The vial attachment portion **514** may comprise a flexible skirt, for example configured to circumferentially surround a portion of the vial **550** and/or releasably engage the file (e.g., through a snap fit). Alternatively, or additionally, the vial attachment portion **514** may comprise one or more (e.g., a plurality of) flexible arms, for example configured to releasably engage the vial **550**. The main body **502** may (e.g., also) include a port **512**, for example at an end of the main body **502** opposite the vial attachment portion **514**. The port **512** may be configured to adjustably engage the attachment **524**, as described herein.

[0062] The adjustment member **524** may comprise a cannula **526**, for example configured to extend through the main body **502** of the vial adapter **500** and/or at least partially into the vial **550** for drawing fluid from the vial **550**. The cannula **526** may have a pointed tip, for example to pierce a septum of the vial **550**. Though depicted as a plastic spike, in some examples the cannula **526** may comprise a metal needle and/or other fluid conduit as desired. In some examples, at an end opposite the cannula **526**, the adjustment member **524** may define a port **525**. The port **525** may be configured to receive a device for receiving the liquid transferred from the vial (e.g., a syringe tip, etc.). The adjustment member **524** may comprise an external threading **526**, for example configured to engage complementary threading defined in the port **512** of the main body **502** for adjustably attaching the adjustment member **524** to the main body **502**. By screwing the adjustment member **524** further into or further out of the main body **502**, the cannula **520** may be extended into the vial **550** by variable lengths, for example (e.g., thus) allowing a user to adjust the maximum intended dose to be drawn from the vial **550**. The adjustment member **524** may (e.g., also) include a hub **528**. The hub may be of increased diameter and/or be configured to engage the main body **502**, for example to define the maximum extension of the cannula **520** into the vial **550**.

[0063] With regard to the example of FIG. 5A, the cannula **520** may set to a smaller dose, for example by screwing the adjustment member **524** to the main body **502** to the greatest extent (e.g., until the hub **528** engages the main body **502**). The remaining volume is illustrated in space **522**. FIG. 5B shows the configuration of FIG. 5A, but with a middle dose setting (e.g., see set cannula length at **526**). FIG. 5C shows the configuration of FIG. 5A, but with a large dose setting, as it relates to cannula length, which is shown for example at **528**. FIG. 5D shows a reference image of an initial cannula setting **530** (e.g., before dose selection and with full medicament (e.g., level shown at **42**) in the vial. FIGS. 5B and 5C represent positions where the adjustment member **524** may be incrementally screwed out of the main body **502**, for example to increase the dose capable of being drawn from the vial **550**.

[0064] FIGS. 6A-C illustrate example cannula and/or dose setting aspects generally at **600** that may, for example, be integrated with and/or be in line with the presently described vial adapter. With reference to FIG. 6A, the adapter **610** may include a cannula dose setting member **612**. This is shown again at FIG. 6B in a schematic, with the cannula and/or dose setting member **612** at (e.g., on) a distal portion. In the illustrated example, a tooth **614** on a rotatable member **616** may prevent backwards rotation relative to a fixed member **618** and/or determine the position of the cannula and/or dose, as in FIGS. 4-5D.

[0065] A vial adapter may be configured to prevent the withdrawal of a second dose of fluid from a vial and/or may prevent the attachment of an additional dosing syringe (e.g., a second or third dosing syringe) subsequent to detachment of a first dosing syringe. For instance, the vial adapter

may include a block pin that prevents withdrawal of fluid from the vial after a first dosing syringe is detached from the vial adaptor. The vial adaptor may be configured to incrementally move the block pin downward, toward the vial attachment portion along a plurality of block pin positions, where each block pin position may be based on a position of a ratcheted slider of the vial adaptor. The block pin may be configured to move downward into a pin channel that is coupled to the fluid channel between the syringe attachment portion and the vial attachment portion of the vial adaptor. As such, after the attachment of a predetermined number of syringes to the vial adaptor, the block pin may be moved to the position where the block pin blocks the fluid channel between the vial and the syringe.

[0066] For instance, in some examples, the vial adaptor may be configured to allow a first syringe (e.g., such as a prefilled syringe (PFS) comprising fluid) to be attached to the vial adaptor to transfer fluid into the vial (e.g., such as those used in combination products and vaccines), and a second syringe (e.g., a first dosing syringe) to be attached to the vial adaptor to withdraw a dose of fluid from the vial. During each attachment and/or remove of the respective first and second syringes, the block pin may be incrementally adjusted toward a blocking position where the block pin blocks the fluid channel between the vial and the syringe. After the attachment of a third syringe (e.g., a second dosing syringe), the block pin may be moved into a position where it blocks the fluid channel. This may be done to prevent the accidental withdraw of a second dose of fluid from the vial. Thereafter, when the third syringe is removed, the ratcheted slider may be in a position where it prevents the attachment of an additional syringe (e.g., a fourth syringe or a third dosing syringe). Accordingly, the vial adaptor may be configured to prevent the withdrawal of a second dose of fluid from the vial and/or may be configured to prevent the attachment of a third dosing syringe. As such, the vial adaptor may prevent the withdrawal of a second dose of fluid from the vial and/or may prevent additional syringes from being attached to the vial adaptor, which may reduce or prevent an undesired additional dosing of fluid to a patient.

[0067] FIGS. 7A-G illustrate a vial adapter **700**. Each of FIGS. 7A-7G illustrate the vial adapter **700** in a different state of operation. FIG. 7A illustrates the vial adapter state prior to attaching a syringe (e.g., syringe **110**). The vial adapter **700** may include a body **712** that extends from a proximal end **712a** to a distal end **712b**. The vial adapter **700** may include a syringe connector **710** at the proximal end **712a** of the body, a fluid conduit **714** that extends from the proximal end **712a** to the distal end **712b** of the body, a cannula **716** at the distal end **712b** of the body **712**, and a vial attachment portion **718** at the distal end **712b** of the body **712** for attaching the vial adapter **700** to a vial (e.g., such as vial **550** shown in FIG. 5A, though not shown in FIGS. 7A-7G).

[0068] The cannula **716** may be configured with a pointed tip, for example, that enables the vial adapter **700** to pierce a portion (e.g., a septum) of a vial (e.g., as shown in prior figures). Though depicted as a plastic spike, in other embodiments the cannula **716** may comprise a metal needle or other fluid conduit as desired. Unlike the arms illustrated as the vial attachment portions in FIG. 1, the vial attachment portion **718** is illustrated as a flexible skirt that extends circumferentially about the cannula **716** and receives the vial (e.g., as a snap fit). However, it is contemplated that in other examples the vial attachment portion **718** may be configured as a screw or interference fit.

[0069] A pin channel **744** may include a first end **744a** and a second end **744b**. A block pin **746** may be at least partially disposed within an interior of the pin channel **744**. For example, the block pin **746** may be disposed within the interior of the pin channel **744** proximate the first end **744a**. The block pin **746** may contact (e.g., abut) a ratcheted slider **728**. For example, the block pin **746** may contact (e.g., abut) the ratcheted slider **728** in a first pin position **754**, or may be integrally connected to or monolithically formed with the ratcheted slider **728**. The second end **744b** may extend towards the distal end **712b** of the body and/or the cannula **716**. As with the examples of FIG. 1, additional mechanisms for preventing removal of the vial may be included as part of the vial adapter **700** (e.g., as locking tabs, or by virtue of an interference fit itself, allowing one way attachment, but preventing removal).

[0070] Referring still to FIG. 7A, a first compressible member 720 may be attached to a movable spring 724, for example extending distally into an open cam space 726 that contains the ratcheted slider 728. The ratcheted slider 728 may be disposed within the cam space 726 and/or be axially movable through the cam space 726. The cam space 726 may be disposed axially between the syringe connector 710 and the cannula 716. As depicted, the compressible member 720 may be configured as a compressible bellows configured to form a fluid seal with the walls of the syringe connector 710 throughout compressive movement. The compressible member 720 may be positioned within the syringe connector 710.

[0071] The proximal end of the movable spring 724 may be attached to the compressible member 720 and/or extend distally into the cam space 726. The movable spring 724 may be attached to the compressible member 720, such that distal compression of the compressible member 720 causes the movable spring 724 to translate distally through the cam space 726 for example. Likewise, proximal expansion of the compressible member 720 may cause the movable spring 724 to translate proximally through the cam space 726. The distal end of the movable spring 724 may be configured to engage the ratcheted slider 728. For example, the distal end of the movable spring 724 may be configured to engage teeth 728a of the ratcheted slider 728. Downward movement of the movable spring 724, for example via engagement between the movable spring 724 and a lateral surface 728b of a tooth 728a of the ratcheted slider 728, (e.g., likewise) may force the ratcheted slider 728 to translate downwards within the cam space 726. However, (e.g., subsequent) upward movement of the movable spring 724, for example via upward expansion of the compressible member 720, may cause the distal end of the movable spring 724 to deflect laterally and/or slide over the angled surface of a tooth 728a of the ratcheted slider 728, which may result in the ratcheted slider 728 remaining in its axial position. The distal end of the movable spring 724 may deflect towards the ratcheted slider 728 and/or engage the lateral surface of a tooth 728a proximal to the tooth 728a with which it was previously engaged, for example upon full expansion of the compressible member 720.

[0072] The vial adapter 700 may additionally, or alternatively include a fixed spring 730, for example to prevent reverse movement of the ratcheted slider 728 upon expansion of the compressible member 720 following the downward movement of the ratcheted slider 728. The fixed spring 730 may be disposed within the cam space 726 and/or attached to the body 712. For example, a proximal end of the fixed spring 730 may be fixedly secured to a wall of the cam space 726. When the movable spring 724 moves distally to translate the ratcheted slider 728 distally, the distal end of the fixed spring 730 may deflect laterally, for example to slide over an angled surface of a tooth 728a of the ratcheted slider 728 and/or (e.g., subsequently) deflect towards the ratcheted slider to engage a lateral surface of a subsequent tooth 728a. This engagement between the fixed spring 730 and the ratcheted slider 728 may hold the ratcheted slider 728 axially in place during upward movement of the movable spring 724, for example preventing proximal movement of the ratcheted slider 728. The movable spring 724 and/or fixed spring 730 may be elongate pieces metal or another material that will provide sufficient push and hold for the illustrated ratcheted slider 728, while still allowing for lateral deflection via deformation.

[0073] As depicted, the ratcheted slider 728 may define a plurality of ratchet positions (e.g., four ratchet positions) for the movable spring 724 and/or the fixed spring 730 to engage the ratcheted slider 728. Each ratchet position may include (e.g., be defined by) a distinct tooth 728a of the ratcheted slider 728. For example, a tooth 728a may include a lateral surface of a tooth 728a of the ratcheted slider 728. The teeth 728a of the ratcheted slider 728 may be positioned axially along the ratcheted slider 728. In one example, the ratcheted slider 728 may include (e.g., may define) one or more of, from distal-most position to proximal-most position, a lowest ratchet position 732, a second lowest ratchet position 734, a second ratchet position 738, and a first ratchet position 742. However, it is contemplated that in other embodiments the ratcheted slider 728 can have more or less ratchet positions than shown presently, for example as dictated by the intended number of

syringe connections for a particular implementation.

[0074] FIG. 7B illustrates the vial adapter **700** in a state with a syringe (e.g., syringe **110** in FIG. 1) attached (e.g., though for convenience, the syringe is not illustrated, though the compressible member **720** is shown compressed as if the syringe were compressing it). The attachment of the syringe to the vial adapter **700** may be by one or more of interference fit, snap on, screw on, Luer lock, etc., for example between the syringe and the syringe connector **710** at this proximal portion. [0075] The vial adapter **700** may include a fluid conduit **714**, for example that fluidly connects the syringe connector **710** to the cannula **716**. The fluid conduit **714** may include one or more of an output aperture **714a** open to the syringe connector **710**, a lateral extension **714b** extending from the output aperture **714a** to the axial extension **714c**, an axial extension **714c**, and an input extension **714d** that connects the cannula **716** to the axial extension. The axial extension **714c** of the fluid conduit **714** may be laterally offset from the cam space **726** and/or the cannula **716**, for example where the axial extension **714c** comprises a majority of the length of the fluid conduit **714**. However, it is contemplated that in other embodiments, the cannula **716** (e.g., along with vial attachment portion **718**) may be repositioned, for example so as to be coaxial with the cam space **726** and/or the axial extension **714c** of the fluid conduit **714**.

[0076] In a first (e.g., initial) position, the compressible member **720** may block the output aperture **714a**, thus for example preventing fluid flow between the cannula **716** and the syringe connector **710**. However, connection of a syringe to the syringe connector **710** may compress the compressible member **720** distally such that, for example the output aperture **714a** is exposed and in fluid communication with the syringe. Fluid may be (e.g., therefore) permitted to flow between the vial and syringe through the cannula **716** and fluid conduit **714**. The pin channel **744** may be fluidly connected to the cannula **716** and/or the fluid conduit **714**. For example, the pin channel **744** may be connected to the input extension **714d**. The compressible member **720** may expand to its first (e.g., initial) position, thus again blocking the output aperture **714a**, for example upon syringe removal from the vial adapter **700**. In some examples a syringe may be attached to inject a diluent or excipient into the fluid conduit, through the cannula, and into a vial containing a medicament. Subsequently, a second syringe may be attached to withdraw (e.g., diluted or reconstituted) medicament through the cannula and fluid conduit and into the second syringe. The second syringe may be removed from the vial adapter for use on a user.

[0077] As shown in FIG. 7A, in a first (e.g., initial) state with the compressible member **720** in an expanded state, the movable spring **724** may engage the ratcheted slider **728** in the lowest ratchet position **732**, for example while the fixed spring **730** may engage the ratcheted slider **728** in the lowest ratchet position **732**. As shown in FIG. 7B, attachment of a first syringe to the vial adapter **700** may cause the compressible member **720** to compress, for example moving the ratcheted slider **728** distally through displacement of the movable spring **724**. This may cause the fixed spring **730** to deflect laterally, for example over the second bottommost tooth **728a** of the ratcheted slider **728** and/or move from the lowest ratchet position **732** to the second lowest ratchet position **734**. As stated previously, for example as in FIG. 7B, the compressible member **720** may be compressed such that the output aperture **714a** is exposed to the syringe connector **710** and/or the first syringe, for example allowing for fluid flow from the cannula **716**, through the fluid conduit **714**, and to the first syringe. The block pin **746** may move downward (e.g., toward the second end **744b** of the pin channel **744**), for example pushed downward by the ratcheted slider **728**. The block pin **746** may enter the interior of the pin channel **744**, for example at a second pin position **752**. Alternatively, the block pin **746** may initially reside at least partially within the pin channel **744**, and be further disposed within the pin channel **744** at second pin position **752**.

[0078] FIG. 7C shows a third configuration of the vial adapter **700**, with the first syringe removed. As noted, the compressible member **720** may be in an expanded state (e.g., no syringe attached, and in some examples blocking fluid flow through fluid conduit **714** via output aperture **714a**). In this state, the fixed spring **730** may be (e.g., still) in the second lowest ratchet position **734**, but, for

example the movable spring **724** may be retracted to the second lowest ratchet position **734** from the lowest ratchet position **732**. The movable spring **724** may be retracted to the second lowest ratchet position **734** from the lowest ratchet position **732** by virtue of the holding power of fixed spring **730** on the ratcheted slider **728**. The first syringe may be a prefilled syringe containing a diluent or excipient. The block pin **746** may remain in the second pin position **752**.

[0079] FIG. 7D shows a fourth position of the vial adapter **700**, with a second syringe attached (as with FIG. 7B, above) to the syringe connector **710**. The second syringe may be a first dosing syringe. The first dosing syringe may be attached at the syringe connector and subsequently used to withdraw fluid from the vial. The ratcheted slider **728** may advance (e.g., further advance) downward in response to the second syringe being attached to the syringe connector **710**. For example, the ratcheted slider **728** may advance downward in response to the downward movement of the moveable spring **724** that occurs when the second syringe is attached to the syringe connector **710**. The advancement of the ratcheted slider **728** may cause the fixed spring **730** to move into the second ratchet position **738** of the ratcheted slider **734**. The movable spring **724** may be (e.g., still) in the second lowest ratchet position **734**. Additionally, or alternatively, (e.g., as herein), in some examples, compression of the compressible member **720** may unblock the output aperture **714a** and/or allow fluid flow to the second syringe via fluid conduit **714**. The block pin **746** may move (e.g., further) downward (e.g., further toward the second end **744b** of the pin channel **744**), for example pushed downward by the ratcheted slider **728**. The block pin **746** may move further downward the interior of the pin channel **744**, for example to a second lowest pin position **750**. For example, the block pin may iteratively move downward in the pin channel toward the vial attachment portion. Since the block pin is contacted by the ratcheted slider, which also moves downward toward the vial attachment portion, the ratcheted slider may push the block pin a distance (e.g., equal to the distance the ratcheted slider moves) with each iterative movement of the ratcheted slider. In a lowest position, the block pin may block the input extension and/or the cannula. When the input extension and/or cannula is blocked by the block pin, fluid may not be drawn into the fluid conduit. This prevents an unintended dosage by a user.

[0080] FIG. 7E illustrates an additional state after removal of the second syringe (e.g., first dosing syringe) from the vial adapter **700**. The fixed spring **730** may remain at the second ratchet position **738**, and/or the movable spring **724** may (e.g., now) be at the second ratchet position **738**. Additionally, or alternatively, as with FIG. 7C, the compressible member (e.g., and/or safety valve) **720** may be in an expanded state (e.g., no syringe attached, and in some examples blocking fluid flow via output aperture **714a**). The compressible member **720** (e.g., likewise) may be prevented from compressing, and/or may block the outlet aperture **714a** and fluid flow to the syringe connector **710** from the fluid conduit **714**. The block pin **746** may remain in the second lowest pin position **750**.

[0081] FIG. 7F shows a sixth position of the vial adapter **700**, with a third syringe attached (as with FIGS. 7B and 7D, above) to the syringe connector **710**. The third syringe may be a second dosing syringe. In some examples, in order to avoid an unintended dose, the third syringe may not be able to withdraw fluid (e.g., an additional dose) from the vial via the vial adapter. The block pin may advance to a lowest position with each movement of the ratcheted slider until the block pin blocks the input extension and/or the cannula. When the input extension and/or cannula is blocked by the block pin, fluid may not be drawn into the fluid conduit, which avoids an unintentional dosage. Additionally, or alternatively, the outlet aperture may be blocked, for example by the compressible member, which also may prevent withdraw of fluid from the vial. The position of the ratcheted slider, the compressible member, and/or the block pin may make a second dosage withdraw from the vial impossible. This may be for user safety, as a user may otherwise mistakenly withdraw a second dose.

[0082] Attachment of the second syringe may (e.g., further) advance the ratcheted slider **728** via downward movement of the movable spring **724**, for example such that the fixed spring **730** moves

into the first ratchet position **742** of the ratcheted slider **734**. The movable spring **724** may be (e.g., still) in the second ratchet position **738**. Additionally, or alternatively, (e.g., as herein), in some examples, compression of the compressible member **720** may unblock the output aperture **714a** and/or allow fluid flow to the second syringe via fluid conduit **714**. The block pin **746** may move (e.g., further) downward (e.g., further toward the second end **744b** of the pin channel **744**), for example pushed downward by the ratcheted slider **728**. The block pin **746** may move further downward the interior of the pin channel **744**, for example to the second end **744b** of the pin channel **744**. For example, the block pin **746** may move to a lowest pin position **748**. In the lowest pin position **748** the block pin **746** may block the input extension **714d** that connects the cannula **716** to the fluid conduit **714**. In the lowest pin position **748** the block pin **746** may (e.g., permanently) block fluid flow from entering the input extension **714d**.

[0083] FIG. 7G illustrates an additional state after removal of the third syringe from the vial adapter **700**. Fixed spring **730** may remain at the first ratchet position **742**, and/or the movable spring may (e.g., now) be at the first ratchet position **742**. Additionally, or alternatively, as with FIGS. 7C and 7E, compressible member (e.g., and/or safety valve) **720** may be in an expanded state (e.g., no syringe attached, and in some examples blocking fluid flow via output aperture **714a**). At this point, the vial adapter **700** may be prevented from use with a fourth syringe, for example since in this position the distal end of the ratcheted slider **728** engages a distal end of the cam space **726** which may prevent further downward movement of the ratcheted slider **728**. As a result, the movable spring **724** may be prevented from moving distally via engagement with the ratcheted slider **728**. The compressible member **720** (e.g., likewise) may be prevented from compressing, and/or may (e.g., permanently) block the outlet aperture **714a** and fluid flow to the syringe connector **710** from the fluid conduit **714**. The block pin **746** may remain in the lowest pin position **748**.

[0084] With further reference to FIGS. 7A-G, an example method of using the vial adapter **700** may include attaching a first syringe, for example resulting in moving the compressible member **720** down. Additionally, or alternatively, attaching a first syringe may result in the ratcheted slider **728** pushing the fixed spring **730** into a locked position during use (e.g., as at FIGS. 7A-7B, into locked position at **734**). This may (e.g., then) open up the fluid conduit, for example for transfer of fluid (e.g., diluent or excipient) from the first syringe into the fluid conduit **714**. At FIG. 7C, the first syringe may be removed, and the movable spring **724** may slide up to the second ratchet position **738** (e.g., due to the lack of presence of a syringe).

[0085] At FIG. 7D, a second syringe may be inserted, for example causing sliding of the ratchet **728** downwards and/or (e.g., also) moving the fixed spring **730** to the third ratchet position **740**. Alternatively, or additionally, insertion of a second syringe may also result in opening the fluid flow through conduit **714**. At this point, medicament may be withdrawn into the second syringe, and/or removed from the vial adapter **700** (e.g., resulting in the configuration shown in FIG. 7E). At FIG. 7E, the vial adapter **700** may be locked out by virtue of the position of the ratcheted slider **728** within the cam space **726** and/or may not accept a third syringe in some examples. At FIG. 7F, the vial adapter **700** may accept a third syringe, however, for example the third syringe may be unable to extract fluid as the outlet aperture **714a** is blocked by the block pin **746**. At FIG. 7G the third syringe may not be removable from the syringe connector **710**. For example, the outlet aperture **714a** may be blocked by the block pin **746**. The compressible member **720** (e.g., likewise) may be prevented from compressing, and/or may (e.g., permanently) block the outlet aperture **714a** and fluid flow to the syringe connector **710** from the fluid conduit **714**.

[0086] As discussed herein, the present presents an apparatus, system and method that provides a vial adapter which may be configured to limit the number of times a syringe can be used with an adapter and/or the dose withdrawn through that adapter.

[0087] It should be emphasized that the above-described embodiments of the present invention, particularly, any detailed discussion of particular examples are merely possible examples of

implementations and are set forth for a clear understanding of the principles of the invention. Many variations and modifications may be made to the above-described embodiment(s) of the invention without departing substantially from the spirit and principles of the invention. All such modifications and variations are intended to be included herein within the scope of this disclosure and the present invention and protected by the following claims.

Claims

1. A vial adapter comprising: a body extending from a proximal end to a distal end; a syringe connector proximate the proximal end of the body, the syringe connector configured to engage a syringe; a cannula disposed proximate the distal end of the body, the cannula comprising a pointed tip that is configured to pierce a septum of a vial; a vial attachment portion at the distal end of the body, the vial attachment portion being configured to attach to the vial; a fluid conduit extending from an outlet aperture at the syringe connector to the cannula; and a compressible member disposed within the syringe connector, wherein attachment of a syringe to the syringe connector is configured to axially compress the compressible member from an expanded configuration, in which the compressible member blocks the outlet aperture and prevents fluid flow between the fluid conduit and the syringe connector, to a compressed configuration, in which the outlet aperture is exposed such that the fluid conduit is in fluid communication with the syringe connector.
2. The vial adapter of claim 1, further comprising: a cam space disposed axially between the syringe connector and the cannula; a ratcheted slider disposed within the cam space and axially movable through the cam space, the ratcheted slider comprising a plurality of teeth; a movable spring attached to the compressible member and extending into the cam space, a distal end of the movable spring configured to engage one or more of the plurality of teeth to move the ratcheted slider through the cam space; and a fixed spring disposed within the cam space and attached to the body, a distal end of the fixed spring configured to engage one or more of the plurality of teeth to prevent proximal movement of the ratcheted slider, wherein engagement between the ratcheted slider and a distal end of the cam space prevents compression of the compressible member.
3. The vial adapter of claim 2, wherein each tooth of the plurality of teeth defines one of a plurality of ratchet positions for engagement of the movable spring or the fixed spring, the plurality of positions comprising a lowest ratchet position, a second lowest ratchet position proximal to the lowest ratchet position, a third ratchet position proximal to the second lowest ratchet position, a second ratchet position proximal to the third ratchet position, and a first ratchet position proximal to the second ratchet position.
4. The vial adapter of claim 3, wherein the ratcheted slider is configured to engage the distal end of the cam space when the movable spring is in the first ratchet position and the fixed spring is in the third ratchet position.
5. The vial adapter of claim 4, wherein in an initial state the movable spring engages the ratcheted slider in the third ratchet position and the fixed spring engages the ratcheted slider in the lowest ratchet position, wherein attachment of a first syringe to the syringe connector compresses the compressible member and moves the movable spring distally through the cam space, the movable spring engaging the ratcheted slider to move the ratcheted slider downwards within the cam space such that the fixed spring moves from the lowest ratchet position to the second lowest ratchet position, wherein subsequent expansion of the compressible member following detachment of the first syringe from the syringe connector moves the movable spring proximally within the cam space, and engagement between the fixed spring and the ratcheted slider prevents proximal movement of the ratcheted slider, such that the fixed spring remains in the second lowest ratchet position and the movable spring moves to the second ratchet position.
6. The vial adapter of claim 5, wherein attachment of a second syringe to the syringe connector compresses the compressible member and moves the movable spring distally through the cam

space, the movable spring engaging the ratcheted slider to move the ratcheted slider downwards within the cam space such that the fixed spring moves from the second lowest ratchet position to the third ratchet position, wherein subsequent expansion of the compressible member following detachment of the second syringe from the syringe connector moves the movable spring proximally within the cam space, and engagement between the fixed spring and the ratcheted slider prevents proximal movement of the ratcheted slider, such that the fixed spring remains in the third ratchet position and the movable spring moves to the first ratchet position, wherein the ratcheted slider engages the distal end of the cam space when the movable spring is in the first ratchet position and the fixed spring is in the third ratchet position.

7. The vial adapter of claim 1, wherein the compressible member comprises a compressible bellows.

8. The vial adapter of claim 2, wherein the movable spring and the fixed spring comprise metal.

9. The vial adapter of claim 2, wherein the cam space extends axially along an axis that is laterally offset from a central axis of the cannula.

10. The vial adapter of claim 1, wherein the vial attachment portion comprises a flexible skirt extending circumferentially about the cannula.

11. A vial adapter comprising: a body defining a proximal end and a distal end; a fluid conduit extending through the body from the proximal end to the distal end, wherein the body defines a circumferential wall at a distal end of the fluid conduit; a cannula disposed at the distal end of the body, the cannula comprising a pointed tip that is configured to puncture a septum of a vial; a vial attachment portion disposed about the fluid conduit and configured to engage the vial; and a gasket disposed within and configured to move axially through the fluid conduit, the gasket comprising a syringe connector configured to engage a syringe and at least one aperture opposite from and in fluid communication with the syringe connector, the aperture being radially offset from a center of the gasket, wherein the gasket is configured to move through the fluid conduit to a distalmost position, wherein in the distalmost position the gasket engages the circumferential wall so as to block the aperture and prevent fluid flow between the cannula and the syringe connector.

12. The vial adapter of claim 11, wherein the gasket comprises a cam member positioned within a cam track, the cam track disposed within the fluid conduit, and the vial adapter further comprises: a spring positioned within the fluid conduit and engaged with the gasket so as to bias the gasket towards the proximal end of the body, wherein the cam track is configured to engage the cam member to control movement of the gasket through the fluid conduit through a plurality of positions, the plurality of positions comprising: a first position; a second position distal to the first position, wherein the gasket is configured to move from the first position to the second position upon attachment of the syringe to the gasket, wherein fluid can be transferred between the vial and the syringe in the second position; and a third position distal to the first position and proximal to the second position, wherein the spring is configured to at least partially move the gasket from the second position to the third position upon disconnection of the syringe from the gasket.

13. The vial adapter of claim 12, wherein the syringe is a first syringe, and the cam track is configured to engage the cam member to further control movement of the gasket through the fluid conduit through the plurality of positions, the plurality of positions further comprising: a fourth position distal to the second and third positions, wherein the gasket is configured to move from the third position to the fourth position upon attachment of a second syringe to the gasket, wherein fluid can be transferred between the vial and the syringe in the fourth position; a fifth position proximal to the fourth position and distal to the third position, wherein the spring is configured to at least partially move the gasket from the fourth position to the fifth position upon disconnection of the second syringe from the gasket; and a sixth position distal to the fourth and fifth positions, wherein the gasket is configured to move from the fifth position to the sixth position upon attachment of a third syringe to the gasket, wherein the gasket engages the circumferential wall in the sixth position.

14. The vial adapter of claim 12, wherein the spring is a coil spring.
 15. The vial adapter of claim 14, wherein the spring is positioned coaxially with the cannula.
 16. The vial adapter of claim 11, wherein the body is comprised of a first material, and the gasket is comprised of a second material, the second material different than the first material.
 17. The vial adapter of claim 16, wherein the body comprises plastic, and the gasket comprises silicone.
 18. The vial adapter of claim 11, wherein the vial attachment portion comprises a flexible skirt extending circumferentially about the cannula.
 19. The vial adapter of claim 11, wherein the aperture comprises a first aperture and second aperture, the first aperture radially spaced from the second aperture.
 20. The vial adapter of claim 11, wherein the aperture comprises a single aperture extending circumferentially about the cannula.
 21. A vial adapter comprising: a main body comprising: a vial attachment portion configured to engage a vial; a port defining an internal thread; and an attachment portion, comprising: a cannula configured to extend through the main body and at least partially into the vial; and an external thread configured to engage with the internal thread, such that engagement of the external thread and the internal thread secures the attachment portion to the main body, wherein incremental advancement of the external thread along the internal thread via rotation of the attachment portion adjusts a distance the cannula extends into the vial.
 22. A vial adapter comprising: a body extending from a proximal end to a distal end; a syringe connector proximate the proximal end of the body, the syringe connector configured to engage a syringe; a cannula disposed proximate the distal end of the body, the cannula comprising a pointed tip that is configured to pierce a septum of a vial; a vial attachment portion at the distal end of the body, the vial attachment portion being configured to attach to the vial; a fluid conduit extending from an outlet aperture at the syringe connector to the cannula; a pin channel comprising an interior, the interior fluidly connected to the fluid conduit; a block pin configured to move within the interior; and a ratcheted slider configured to contact the block pin, the ratcheted slider configured to axially compress the block pin to a position in which the block pin blocks a fluid connection between the fluid conduit and the cannula.
 23. A method of using a vial adapter, the vial adapter comprising a vial attachment portion, a cannula, a syringe connector, a ratcheted slider, a movable spring, a compressible member, and a fluid conduit, wherein the method comprises: attaching a vial to the vial attachment portion, wherein a septum of the vial is punctured by a pointed tip of the cannula; attaching a syringe to the syringe connector, wherein attachment of the syringe to the syringe connector causes the compressible member to translate distally, the distal translation of the compressible member causes the movable spring to translate distally, and the distal translation of the movable spring causes the ratcheted slider to translate distally; and transferring a fluid between the vial and the syringe, wherein the fluid is transferred through the cannula and a fluid conduit, and wherein the fluid conduit extends from an outlet aperture at the syringe connector to the cannula.
 24. The method of claim 23, wherein attachment of the syringe to the syringe connector is configured to axially compress the compressible member from an expanded configuration, in which the compressible member blocks the outlet aperture and prevents fluid flow between the fluid conduit and the syringe connector, to a compressed configuration, in which the outlet aperture is exposed such that the fluid conduit is in fluid communication with the syringe connector.
 25. The method of claim 23, wherein the vial adapter further comprises a block pin configured to move within a pin channel, and wherein the ratcheted slider is configured to contact the block pin and to axially compress the block pin to a position in which the block pin blocks a fluid connection between the fluid conduit and the cannula.
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